

Benchmarking Large Language Models for Lemmatization and Translation of Finnic Runosongs

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<https://github.com/hsci-r/filter-llm-lemmatization>

Interlinked projects in Estonia and Finland

FILTER consortium: Formulaic intertextuality, thematic networks and poetic variation across regional cultures of Finnic oral poetry,
(Kati Kallio & Eetu Mäkelä, Finnish Literature Society & University of Helsinki with Estonian Literary Museum, Research Council, 2020–2024)

A corpus-based approach to folkloric variation: regional styles, thematic networks, and communicative modes in runosong tradition
(Mari Väina, Estonian Literary Museum, PRG 1288, 2021–2025)

Estonian Roots: Centre of Excellence for transdisciplinary studies on ethnogenesis and cultural diversity
(Prof. Kristiina Tambets, 2024–2030; Mari Väina leading the part on folklore)

Variation of content and form of Finnic oral poetry in relation to linguistic and ethnic histories (IKAKE)
(Kati Kallio, Kone Foundation; Finnish Literature Society 2024–2028)

Runosongs

- versatile oral tradition shared by most Finnic languages (South and North Estonian, Votic, Ingrian, Karelian, Ludic and Finnish)
- digital corpus c. 250 000 texts recorded 1564–, mostly 1800–1960
- lots of linguistic (dialectal, morphological, orthographic), poetic, and content variation
- 1 084 389 different word forms

Map from Riho Grünthal (2020): The spread zones and contacts of medieval Finnic in the Northeastern Baltic Sea area: Implications for the rate of language change, *Journal of Historical Sociolinguistics* 2020; 6(2).



Non-standard, dialectal, poetic and orthographical variation

- Finnic languages have rich morphological variation
- non-standard local oral dialects and under-resourced minority languages
- complex linguistic distributions
 - blurry borders between dialects and languages
 - multilingual areas and songs
 - oral poetry may carry archaic words and expressions or features from close linguistic areas
 - also word forms that appear only in historical poetic context
- no dictionaries, parsers or NLP tools covering the whole data
- the ultimate goals (from folklorists' point of view) is to reveal **recurrent units** (motives)

Large language models

- Task: to add layers (translation, grammatical analysis, etc) to the whole corpus of 250000 texts to open new outlooks for analysis and to handle variability

*Kuz päi kellozet kuuluu,
sielt päi sulhazet tuloo.*

LLM prompt

Orig.	English	Norm.	Lemma	Etym. root
kuz	six	kuusi	kuusi	*kuuci
päi	days	päiviä	päivä	*päivä
kellozet	bells	kelloset	kello	*kello
kuuluu	are heard	kuuluu	kuulua	*kuuludak
sielt	from there	sieltä	se	*se
päi	direction	päin	pää	*pää
sulhazet	grooms	sulhaset	sulhanen	*sulhanen
tuloo	come	tulevat	tulla	*tulldak

deepseek-R1

Test set: manual annotation

- Random sampling of ~200 heterogeneous texts
 - language and dialect are not coded in database → approximate sampling by group of parishes
 - sampling data by centuries (17th–18th, 19th–20th)
- For each poem: dialect, orthography and genre
- For each word:
 - Normalized orthography
 - Lemma in original dialect
 - Lemma in standard Estonian or Finnish (if available)
 - Etymological root
 - Translation into English
- These fields allow search and grouping material on different levels, from surface forms to semantics

Test set: manual annotation

	#texts	#verses	#words
North Estonian	62	1113	4213
South Estonian	30	530	2093
<1800 Finnish	63	982	2973
>=1800 Finnish	25	235	764
Karelian	17	636	2030
Ingrian	8	272	893
Votic	6	144	418
Ludic	4	30	109
Swedish	1	4	20
Total	216	3946	13513

Publicly available: <https://github.com/hsci-r/filter-llm-lemmatization>

Prompt engineering is not a profession



- Collective creative process
- Prompts include
 - language details
 - cultural context
 - output format
 - concrete instructions
 - intermediate steps

Prompt pipelines

```
{
  "system": "system/main_system.txt",
  "steps": [
    {
      "name": "table_only",
      "task_prompts": [
        "context/general_{lang}.txt",
        "context/cultural.txt",
        "context/poetic.txt",
        "context/linguistic_{lang}.txt",
        "context/phono_{lang}.txt",
        "format/table_format.txt",
        "task/table_{lang}.txt",
        "input/input.txt"
      ],
      "chunking": {
        "chunk_notice": "connectors/
        chunk_notice.txt"
      },
      "validation": {
        "enforce_first_column": true,
        "min_table_cols": 7
      }
    }
  ]
}
```

- 1 **Table only:** Model performs direct word-by-word table analysis.
 - 2 **Translation + table (merged):** Model first translates the runosong and immediately outputs the analysis table; tests whether translation aids understanding.
 - 3 **Translation → table (sequential):** Two calls — first translation, then use that output as input for table creation.
 - 4 **Translation → fix → table:** Adds an intermediate correction stage to refine the translation table before analysis.
 - 5 **Translation → fix → table → fix:** Adds a final self-correction step to improve consistency in the produced table.
- each pipeline in two variants: with and without context

Table-only prompt without context

----- 1. SYSTEM -----

You are an expert in Finnic runosong tradition and historical linguistics with deep knowledge of dialectal variations across Finnish, Estonian, Karelian, Votic, Ingrian (Izhorian, “inkeroisen kieli”, “isuri keel”), Veps, and other Finnic languages. Your task is to understand the text as a whole considering separately each word and its components according to information on and the procedure specified below.

----- 2. USER -----

Create a single word-by-word analysis table with this format:
original form	comment	English translation	normalized orthography	lemma (original)	lemma (modern)	etymological root
[Word as in text]	[Translation notes]	[English equivalent]	[Modern spelling]	[Basic form in original]	[Modern language lemma]	[Etymological root(s)]

Analyze the provided Finnic runosong text and translation to create a comprehensive word-by-word analysis table.

ANALYSIS GUIDELINES

For each word in the original text:
- The "original form" column should use the exact word from the original language text
- Add helpful comments about interpretation challenges or linguistic features in the "comment" column

REFRAINS
- Detect if the song contains refrain words at the end or in the middle of each line: do not analyse these words, mark these as [refrain].
- Detect if the song contains verse-length refrains: do not analyse these words, mark these as [refrain].

- In "English translation," provide the best English equivalent for this specific word based on the translation - main word form that can serve as a keyword entry in English dictionary (nominative singular, present tense infinitive with to, no prefixes nor modalities), give only translation of the main word form, do NOT add information what is given with morphological endings
- For "lemma (original)," provide the basic form in the dialect of the text (for nouns: nominative singular, for verbs: present tense infinitive, use forms with -ma or -me or -mä ending & NOT -da/-ta ending) with original phonological peculiarities, with derivational suffixes, without morphological endings
- For "lemma (modern)," give the the equivalent in standard phonology, that can serve as a keyword entry to standard Estonian dictionary with derivational suffixes, without morphological endings, do NOT separate compounds into parts, do NOT change etymological stems.
- For "etymological stem," include the stem word(s), i.e. main word forms (in case of verbs, ma-infinitive NOT da-infinitive) that would serve as dictionary entries in ETY, from the eki.ee site), remove derivational suffixes, separate multiple possibilities with commas and compound components with &

IMPORTANT NOTES

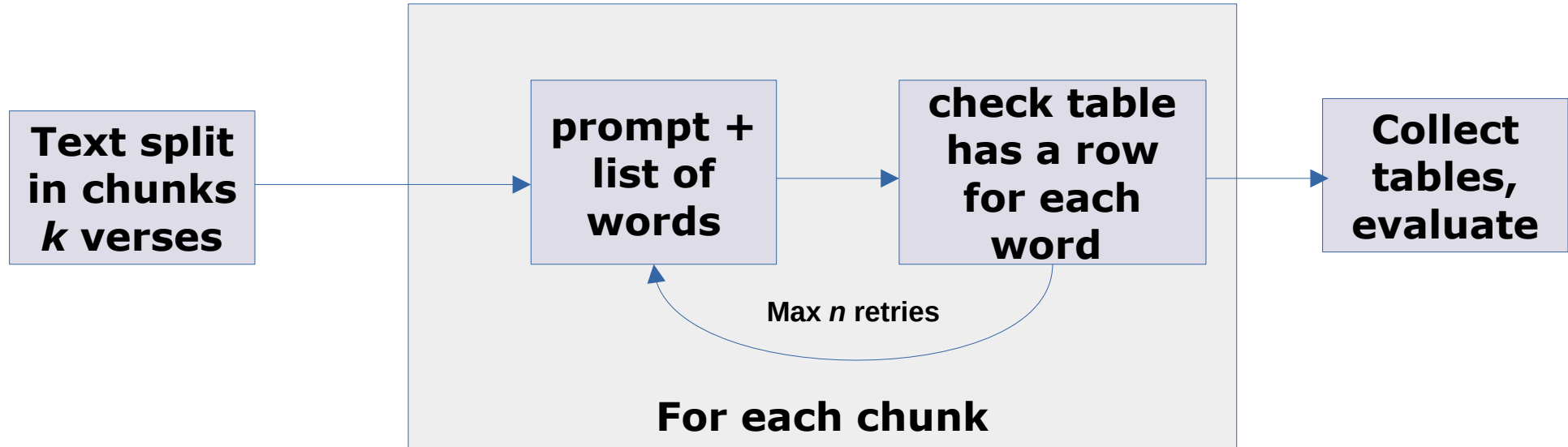
1. Focus ONLY on creating the required tables. Do not include any other tables, dialect identification, or additional commentary.
2. Include all words from the original text in your analysis, even particles and function words.
3. Be consistent in your analysis throughout the table.
4. Use the provided translation to inform your analysis, but focus on an accurate word-by-word breakdown.
5. For words that are difficult to analyze, provide your best linguistic assessment based on context and comparative Finnic linguistics.

Begin your analysis by examining the original text and its translation, then create the word-by-word analysis table as specified above.

Original text:
kost sa orav tee siia osasid tulla
siia kõrgeella katussella
siia uhkeella hoonessa
kost sa siilu silma võtid
kost sa nirgi nina võtid

I expect the first column in the table to contain all these words, in this order: ['kost', 'sa', 'orav', 'tee', 'siia', 'osasid', 'tulla', 'siia', 'kõrgeella', 'katussella', 'siia', 'uhkeella', 'hooness', 'a', 'kost', 'sa', 'siilu', 'silma', 'võtid', 'kost', 'sa', 'nirgi', 'nina', 'võtid'] All columns should be filled for every word.

Processing



Models

Label	HuggingFace path / source	number of parameters	chunk size	maximum output
copy-word	-	-	-	-
poro	LumiOpen/Llama-Poro-2-70B-Instruct	70B	25	4000
llama	meta-llama/Llama-3.3-70B-Instruct	70B	100	16000
databricks	databricks/dbrx-instruct	132B	25	8000
mixtral	mistralai/Mixtral-8x22B-Instruct-v0.1	141B	100	8000
deepseek	unsloth/DeepSeek-R1-BF16	671B	100	16000
claude	Claude-3.7-Sonnet-20250219	unknown	25	8000

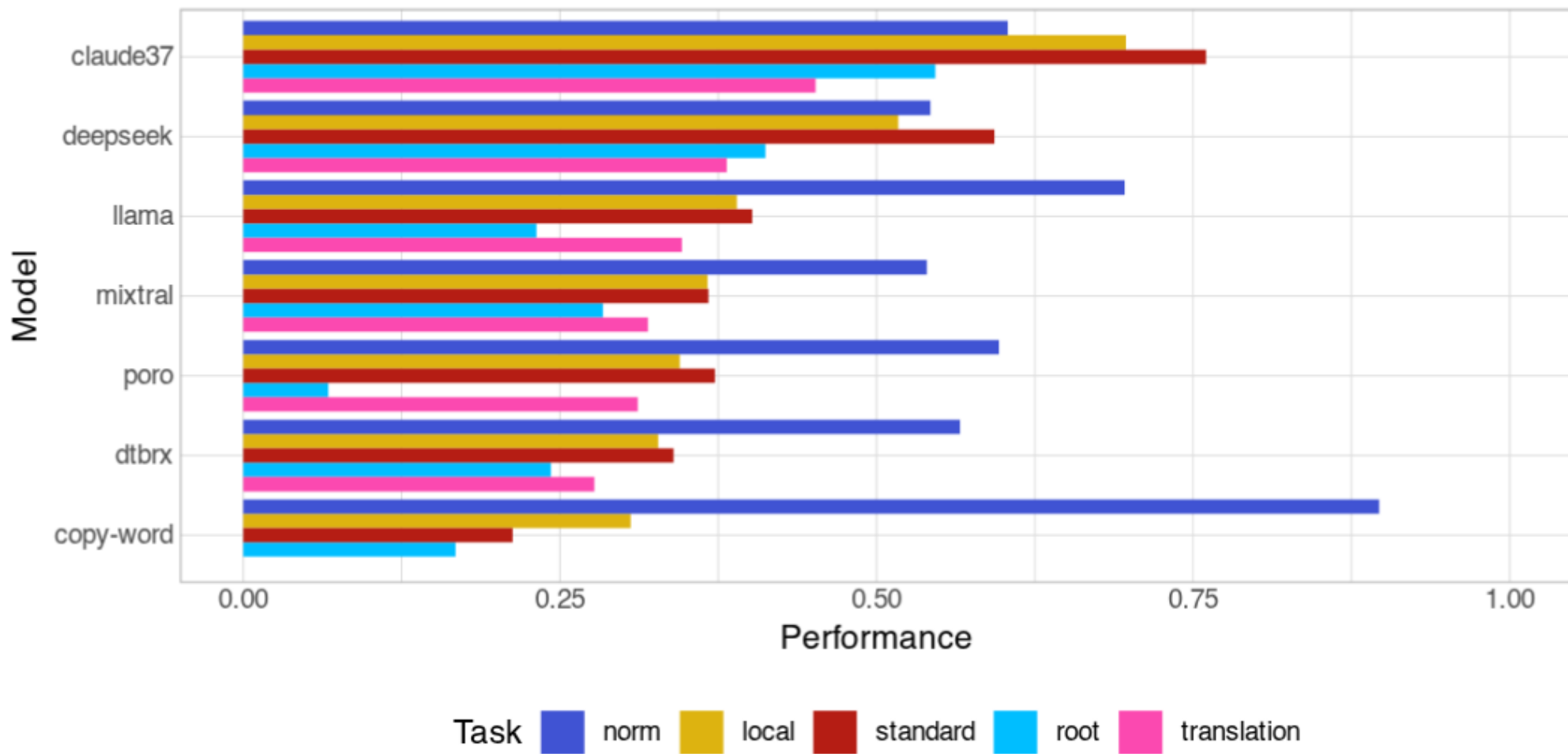
Evaluation

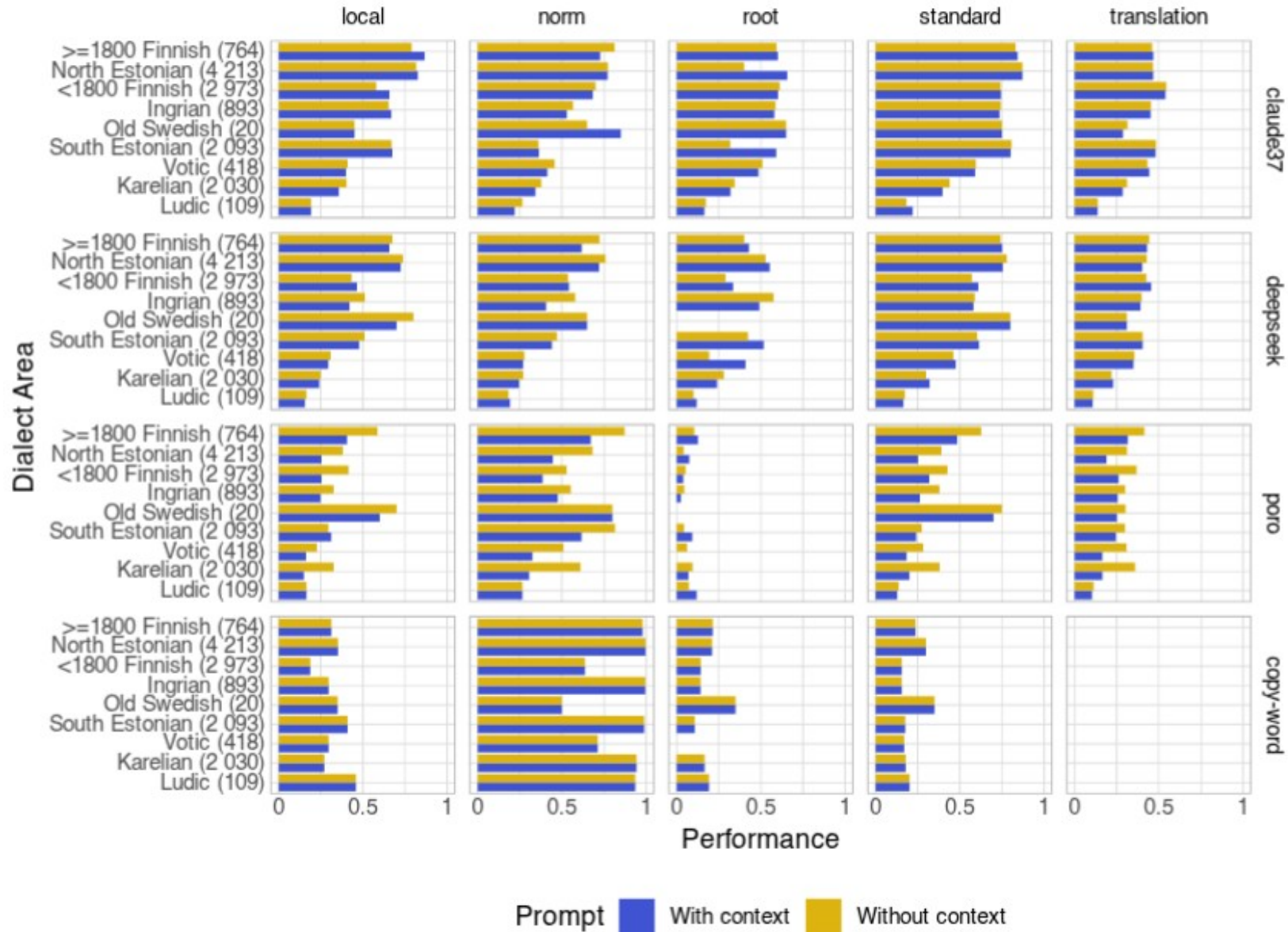
- Exact match (% of words)
- Distance:
 - orphography, lemma and root → similarity based on Levenshtein distance
$$\text{similarity} = 1 - \text{dist}(s_1, s_2) / (|s_1| + |s_2|)$$
 - translation → cosine similarity between word embeddings

Results

Outcomes

- Claude is the best, Deepseek is the best free
- Difference between the models is much higher than a difference between pipelines
- The simple "do analysis table" without any translation or fixing performs slightly better than with any added finesse





Error analysis

- Confusing normalization with standardization, i.e. replacing dialectal or minority language forms with modern Finnish or Estonian.
- Substituting with common synonyms rather than producing faithful lemmatization.
- Misinterpreting archaic morphological forms.
- Inconsistent handling of homonymy and dialectal variation: not recognizing dialectal or minority language words and mixing them with their homonyms in the major languages.
- Refusal to analyse obscene or culturally marked content.
- Difficulty with recognizing onomatopoetic or nonsense words, and refrains, i.e. recurrent words with meanings separate from the main text.
- Alternative interpretations and inability to follow guidelines.

Still, ability to understand runosongs is amazing

original verse	comments	English translation
ku olli nuuri neiokõne	South Estonian dialect with diminutive form “neiokõne”	When I was a young maiden
kui ma kasvi kabokõne	“kabokõne” is a diminutive form of “kabo” (maiden, young woman)	When I grew up as a young girl
lätsi marja sis mäe päälõ	“lätsi” is South Estonian past tense form of “minema” (to go)	I went berry-picking on the hill
lätsi orgo ubinahe	“ubinahe” refers to apple orchard (illative case)	I went to the valley to the apple orchard
panni ma tuppõ tuima ravva	“tupp” = sheath, “tuim raud” = cold iron/steel (knife)	I put the cold steel in the sheath
vaivaväidse pannu vüü ala	“vaivaväits” = poor/miserable knife, “vüü ala” = under the belt	I put the poor knife under my belt

Claude 3.7

Future work

- Add dialect recognition step
- Improve prompts
- Add external resources
- Process the whole corpus (~1M tokens) and tackle recurrent units
- Fine-tune models

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